Second order inhomogeneous ODEs

Key Points

To solve a 2nd order differential equation of the form

$$a\frac{d^2y}{dx^2}+b\frac{dy}{dx}+cy=Q(x)$$

Find CF; y_{CF} by solving homogeneous

Find PI; y_{PI} by guessing and substituting to find constants

For general $y=y_{CF}+y_{PI}$

Include all derivates of your guess in your guess and if the initial guess doesn't work then try to multiply by x

Basic Skills

Q1. Find the general solutions to the following homogeneous differential equations

a) $\frac{d^2y}{dx^2}+2\frac{dy}{dx}-10y=0$

b) $\frac{d^2y}{dx^2}-6\frac{dy}{dx}+9y=0$

c) $\frac{d^2y}{dx^2}+9y=0$

Q2. Find the particular solution to $\frac{d^2y}{dx^2}+5\frac{dy}{dx}-14y=0$ when the initial conditions are $y(0)=\frac{dy}{dx}(0)=1$

Q3. Show that $y=3\cos 2x$ solves $\frac{d^2y}{dx^2}+7y=9\cos 2x$

Competency Skills

Q4. For the differential equation $\frac{d^2y}{dx^2}+\frac{dy}{dx}-6y=e^{2x}$

a) Find the complementary function by solving the homogeneous version $\frac{d^2y}{dx^2}-\frac{dy}{dx}+6y=0$

b) For the particular integral try $y=ae^{2x}$ followed by $y=axe^{x}$ and hence write the general solution

c) Given $y(0)=0$ and $\frac{dy}{dx}(0)=5$ find a particular solution to $\frac{d^2y}{dx^2}+\frac{dy}{dx}-6y=e^{2x}$

Q5. Find the general solution to the differential equation $\frac{d^2y}{dx^2}+4\frac{dy}{dx}+4y=4\cos x$

Q6. Find the particular solution to $x\frac{d^2y}{dx^2}+4xy=x^3-x$ given that $y(0)=1$ and $y\left(\frac{\pi}{4}\right)=5$

Q7. Find the general solution to $\frac{d^2y}{dx^2}-10\frac{dy}{dx}+25y=xe^{5x}$

Advanced Skills

Q8. Use the methodology for second order ODEs to find a general solution to the first order ODE $3\frac{dy}{dx}+5y=25-2\sin 3x$

Q9. The charge C in a capacitor at time t is initially zero and the rate of charge is also zero and obeys the second order differential equation $\frac{d^2C}{dt^2}+50\frac{dC}{dt}+5000C=500\cos 50t$

a) Find the charge at any time t

b) Write $C(t)$ in amplitude phase form

c) Sketch the graph for $C(t)$ and describe the behaviour initially and as time goes to infinity

Extension

Q10. Determine what sort of differential equations have a solution which decays to zero as $t\to\infty$

Hint: Consider how the CF and PI affect the general solution